

DTM — Decision Traceability Module

Parent Standard: Operational Decision Integrity Standard (ODIS)

Category: Governance & Enforcement

Subcategory: Decision Traceability

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Compatibility: OOF Methodology OS · Operational Decision Integrity Standard (ODIS) · Operational Evidence & Auditability Standard (OEAS) · Runtime Integrity Standard (RIS) · Operational Context Integrity Standard (OCIS) · Authority & Accountability Layer Standard (AALS) · INTEGROS® — Integrity Standard · Autonomous Runtime Systems · AI Agent Infrastructures

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Decision Traceability Module (DTM) defines the structural conditions under which runtime operational decisions, adaptive decision evolution, distributed decision coordination, and consequence-bearing decision states remain materially reconstructable, reviewable, attributable, and operationally governable across runtime environments.

A system satisfies DTM only if:

- operational decisions remain materially traceable
- runtime decision pathways remain reconstructable
- adaptive decision evolution remains reviewable
- distributed decision coordination preserves attribution continuity
- hidden decision states do not destabilize operational legitimacy

A system that preserves runtime execution while operational decisions materially lose traceability does not satisfy DTM.

Module Operational Role

DTM defines the decision traceability layer of ODIS by preserving reconstructable runtime decision continuity across operational environments.

Module Operational Space

DTM governs the decision traceability space of ODIS by preserving operational decision reconstruction, runtime attribution continuity, adaptive decision reviewability, distributed coordination traceability, and consequence-bearing decision visibility across runtime systems.

Module Function

The module applies wherever systems must preserve:

- operational decision traceability
- runtime decision reconstruction
- adaptive decision reviewability
- distributed attribution continuity
- consequence-bearing decision visibility
- governance-valid decision accountability

Its function is to ensure that runtime operational decisions remain materially reconstructable strongly enough to preserve operational legitimacy across execution environments.

Minimum Implementation Framework

1. Define the Decision Traceability Object

The organization must define which operational decision structures require traceability governance.

This may include:

- autonomous runtime decisions
 - orchestration-level decision routing
 - adaptive decision evolution
 - distributed operational coordination
 - escalation-sensitive runtime choices
 - consequence-bearing operational actions
 - delegated decision pathways
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2. Define Decision Traceability Conditions

The system must define the conditions under which operational decisions remain materially reconstructable and operationally aligned.

This includes:

- runtime decision visibility
 - attribution continuity
 - adaptive decision reviewability
 - distributed coordination traceability
 - governance-valid reconstruction capability
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3. Define Decision Visibility Failure Detection Logic

The system must define how materially weakened decision traceability or hidden decision-state instability is identified.

This may include:

- unreconstructable runtime decisions
 - attribution fragmentation
 - adaptive decision opacity
 - distributed coordination invisibility
 - consequence-bearing hidden decision pathways
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4. Define Operational Response or Governance Logic

The system must define governance logic for materially weakened decision traceability conditions.

Governance response may include:

- reconstruction enforcement
 - attribution synchronization
 - traceability escalation review
 - adaptive decision restriction
 - operational review activation
 - operational invalidation where required
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5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of operational decision continuity and hidden-decision states. A system must not remain decision-valid if runtime operational decisions materially lose traceability while systems continue assuming governance-valid operational legitimacy remains preserved.

Use Case 1 — Distributed AI Decision

Infrastructure

Scenario

A distributed AI environment continuously generates adaptive runtime decisions across orchestration systems and autonomous operational infrastructures.

Application

DTM preserves governance-valid decision reconstruction through runtime attribution governance and adaptive decision reviewability continuity.

Result

The organization gains stronger operational decision accountability and reduced hidden runtime decision opacity across distributed AI systems.

Use Case 2 — Autonomous Operational Control

Environment

Scenario

An autonomous operational infrastructure continuously coordinates realtime decisions across distributed runtime systems and adaptive execution environments.

Application

DTM preserves governance-valid runtime decision visibility through distributed coordination traceability and operational reconstruction governance.

Result

The environment gains stronger runtime accountability and reduced hidden operational decision instability across autonomous operational systems.

Canonical Closing Statement

If runtime operational decisions cannot remain materially reconstructable across execution environments, systems may preserve operational continuity while decision accountability progressively loses governance-valid visibility.

Related Documents

→ [Operational Decision Integrity Standard - \(ODIS\)](#).

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